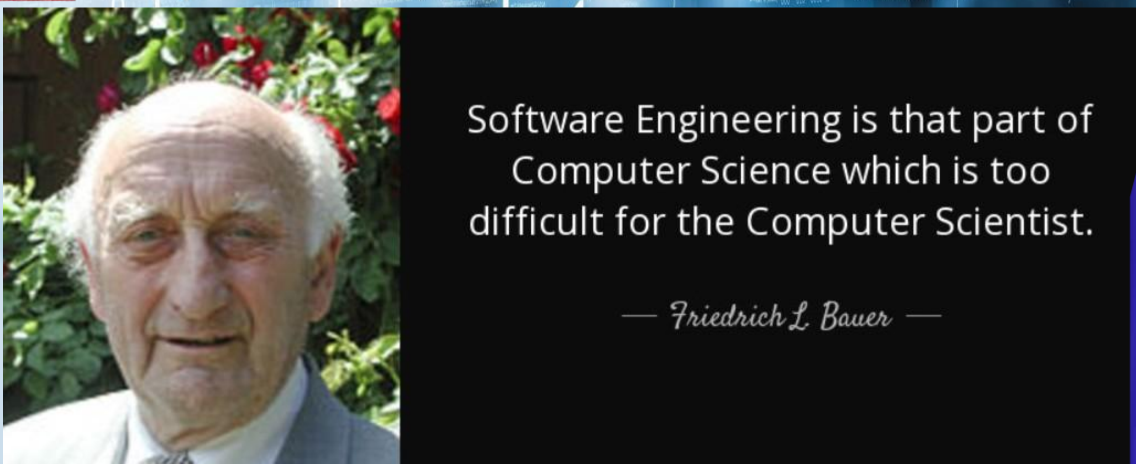
SOFTWARE ENGINEERING

CE-SEM 5

VIVA QUESTIONS



Software Engineering

1. **What is Software Engineering?**
   Software Engineering is the systematic, disciplined, and quantifiable approach to the development, operation, and maintenance of software. It applies engineering principles to software creation.

2. **What is the difference between a Program and Software?**
   A program is a single executable file with a set of instructions. Software is a collection of programs, documentation, and data designed to perform a comprehensive task.

3. **What is a Software Development Life Cycle (SDLC)?**
   SDLC is a structured process for building software that consists of distinct phases: Planning, Analysis, Design, Implementation, Testing, Deployment, and Maintenance.

4. **List the different SDLC models.**
   Common SDLC models include the Waterfall Model, Iterative Model, Spiral Model, V-Model, Agile Model, and Prototyping Model.

5. **Explain the Waterfall Model.**
   The Waterfall Model is a linear and sequential approach where each phase must be completed before the next one begins. It is simple but inflexible to changing requirements.

6. **What is the Agile Methodology?**
   Agile is an iterative and incremental approach to software development that emphasizes flexibility, customer collaboration, and delivering working software in short timeframes called "sprints."

7. **What is a Software Requirement?**
   A software requirement is a description of what a software system should do, the services it provides, and the constraints under which it operates.

8. **Differentiate between SRS and FRS.**
   SRS (Software Requirements Specification) describes what the system will do from a developer's perspective. FRS (Functional Requirements Specification) describes how the system will behave from a user's perspective.

9. **What is Feasibility Study?**
   A feasibility study analyzes the practicality of a proposed software project, considering technical, economic, operational, and legal factors.

10. **What is Software Design?**
    Software Design is the process of defining the architecture, components, interfaces, and other characteristics of a system or component.

11. What is a DFD?

A Data Flow Diagram (DFD) is a graphical representation of the flow of data through a system. It shows processes, data stores, data flows, and external entities.

12. What is an ER Diagram?

An Entity-Relationship Diagram (ERD) is a visual representation of data objects (entities) and their relationships, used in database design.

13. What is Cohesion?

Cohesion measures the strength of the relationship between elements within a single module. High cohesion (performing one focused task) is desirable.

14. What is Coupling?

Coupling measures the interdependence between software modules. Low coupling (minimal dependencies) is desirable.

15. What is Software Testing?

Software Testing is the process of evaluating a software item to identify differences between expected and actual results and to assess features and performance.

16. Differentiate between Verification and Validation.

Verification: "Are we building the product right?" (Checking against specifications). Validation: "Are we building the right product?" (Checking against user needs).

17. What is White-box Testing?

White-box Testing is a method where the tester has knowledge of the internal structure/code of the application and tests the internal logic and paths.

18. What is Black-box Testing?

Black-box Testing is a method where the tester has no knowledge of the internal code and tests the functionality based on requirements.

19. What is Unit Testing?

Unit Testing involves testing individual units or components of the software to ensure they work correctly in isolation.

20. What is Integration Testing?

Integration Testing is the phase where individual software modules are combined and tested as a group to expose faults in their interfaces and interaction.

21. What is Alpha and Beta Testing?

Alpha Testing is conducted by the development team within the organization. Beta Testing is conducted by a limited number of end-users at their site before final release.

22. What is a Test Case?

A Test Case is a set of conditions or variables under which a tester will determine if a system under test satisfies requirements and works correctly.

23. What is Software Maintenance?

Software Maintenance is the process of modifying a software product after delivery to correct faults, improve performance, or adapt to a changed environment.

24. What are the types of Software Maintenance?

The four types are: Corrective (fixing bugs), Adaptive (adapting to new environments), Perfective (improving performance/usability), and Preventive (making it more maintainable).

25. What is Software Configuration Management (SCM)?

SCM is the task of tracking and controlling changes in the software, including version control and baseline management.

26. What is a Use Case Diagram?

A Use Case Diagram is a UML diagram that shows the interactions between users (actors) and the system to achieve specific goals.

27. What is a Class Diagram?

A Class Diagram is a static UML diagram that shows the system's classes, their attributes, methods, and the relationships among objects.

28. What is a Sequence Diagram?

A Sequence Diagram is a UML interaction diagram that shows how objects communicate with each other in terms of a sequence of messages over time.

29. What is Risk Management?

Risk Management is the process of identifying, analyzing, and responding to project risks to minimize their impact on the project's success.

30. What is CMMI?

The Capability Maturity Model Integration (CMMI) is a process level improvement training and appraisal program that provides organizations with the essential elements of effective processes.

31. What is the role of a Project Manager?

A Project Manager plans, executes, and closes projects. They are responsible for scope, time, cost, quality, resources, communication, and risk.

32. What is the difference between Quality Assurance and Quality Control?

QA is process-oriented and focuses on preventing defects. QC is product-oriented and focuses on identifying defects in the finished product.

33. What is COCOMO Model?

The Constructive Cost Model (COCOMO) is a procedural software cost estimation model used to predict the effort and duration of a project.

34. What is the purpose of a Gantt Chart?

A Gantt Chart is a bar chart that illustrates a project schedule, showing task start and end dates, dependencies, and the current progress.

35. What is Refactoring?

Refactoring is the process of restructuring existing code without changing its external behavior to improve its readability, reduce complexity, or make it easier to maintain.

36. What is a Software Prototype?

A Software Prototype is an incomplete but working version of the software, built to understand requirements better and gather early user feedback.

37. What is the Spiral Model?

The Spiral Model combines the iterative nature of prototyping with the controlled and systematic aspects of the waterfall model. It emphasizes risk analysis.

38. What is the V-Model?

The V-Model is an extension of the waterfall model where for every development phase, there is a corresponding testing phase (e.g., Unit Testing pairs with Coding).

39. What is the main principle of Extreme Programming (XP)?

XP emphasizes customer satisfaction, frequent releases in short cycles, teamwork, and improved software quality through practices like pair programming and continuous testing.

40. What is a Story in Agile?

In Agile, a Story is a simple, informal description of a feature told from the perspective of the user. It is the primary unit of work.

41. What is Pair Programming?

Pair Programming is an Agile technique where two programmers work together at one workstation. One writes code (the driver), and the other reviews each line (the navigator).

42. What is the purpose of a PERT Chart?

A PERT (Program Evaluation and Review Technique) Chart is a project management tool used to schedule, organize, and coordinate tasks. It shows the dependencies between tasks.

43. What is Modularization?

Modularization is the design technique of separating the functionality of a program into independent, interchangeable modules.

44. What is a Data Dictionary?

A Data Dictionary is a centralized repository of information about data, such as meaning, relationships to other data, origin, usage, and format.

45. What is the difference between Structural and Behavioral Testing?

Structural Testing (White-box) is based on the internal code structure. Behavioral Testing (Black-box) is based on the external specifications and functionality.

46. What is Cyclomatic Complexity?

Cyclomatic Complexity is a software metric used to indicate the complexity of a program. It measures the number of linearly independent paths through a program's source code.

47. What is Reverse Engineering?

Reverse Engineering is the process of analyzing a software system to extract its design and implementation information, often to recreate the system or parts of it.

48. What is a Walkthrough?

A Walkthrough is a static testing technique where the author of the document explains the content to a team to gather feedback and find defects.

49. What is the role of a Software Metric?

Software Metrics are quantitative measures used to gauge the quality, performance, and progress of a software project (e.g., Lines of Code, Defect Density).

50. What is the significance of the __init__.py file in a Python package from a Software Engineering perspective?

The __init__.py file signifies that a directory is a Python package. From an SE perspective, it enforces modularity, allows for package initialization code, and helps manage namespaces, which are key principles for building large, maintainable software systems.